MapReduce Shape Counting Pipeline Simulation

Coursera Big Data / MapReduce Assignment Submission

STEP 0: HDFS STORE

Partition 1 (6 items)

Square Star
Square Star
Circle Square

Partition 2 (5 items)

Circle Heart
Circle Square
Heart

Partition 3 (5 items)

Triangle Square
Circle Triangle
Triangle

Partition 4 (Given - 5 items)

Square Star
Triangle Star
Square Heart

STEP 1: MAP

Mapper 1

Square Square
Square Star
Star Circle

Mapper 2

Square Circle
Circle Heart
Heart

Mapper 3

Square Circle
Triangle Triangle
Triangle

Mapper 4

Square Square
Star Star
Heart Triangle

STEP 2: SHUFFLE & SORT

Node 1 (Squares & Stars)

Square × 5
Star × 4

Node 2 (Circles & Hearts)

Circle × 4
Heart × 3

Node 3 (Triangles)

Triangle × 5

STEP 3: REDUCE

Key	Value
Square	5
Star	4
Circle	4
Heart	3
Triangle	5

Execution Overview & Explanation

- **Step 0 (HDFS Partitioning):** Distributed the 21 initial elements across 4 balanced data partitions (~5-6 items per partition).
- **Step 1 (Map):** Mapped each shape element into key-value pairs `(key, 1)` and grouped identical keys together locally.
- **Step 2 (Shuffle & Sort):** Transferred and consolidated intermediate pairs from all mappers into 3 reducer nodes by shape key to balance workload.
- **Step 3 (Reduce):** Aggregated key-value counts to calculate the final occurrences: **5 Squares, 4 Stars, 4 Circles, 3 Hearts, and 5 Triangles**.